Understanding the Operating System Booting Process

CSE 4501 : Operating Systems

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1. Introduction

Booting is the process by which a computer system initializes hardware components and loads the OS kernel into memory to begin normal operation. It is the first task that happens when the system is toggled on. It lays the foundation for all subsequent computing tasks.

Key components in the booting process :

- **BIOS (Basic Input/Output System):** Firmware that performs initial hardware checks (POST) and starts the boot loader.
- **Boot Loader:** A small program responsible for loading the OS kernel into memory.
- **Kernel:** The core of the OS that handles resource management and system services after loading.

2.Stages of Booting

2.1 Power-On Self Test (POST)

When the system is powered on:

- The **BIOS/UEFI firmware** performs the POST.
- It checks if essential hardware components (RAM, CPU, keyboard, etc.) are functioning.
- Displays system information and potential error codes (e.g., beep codes).
- If POST passes, the BIOS searches for a bootable device (HDD, SSD, USB, etc.).

This stage is crucial for system integrity and prevents loading the OS in a faulty environment.

2.2 Boot Loader

Once a bootable device is located:

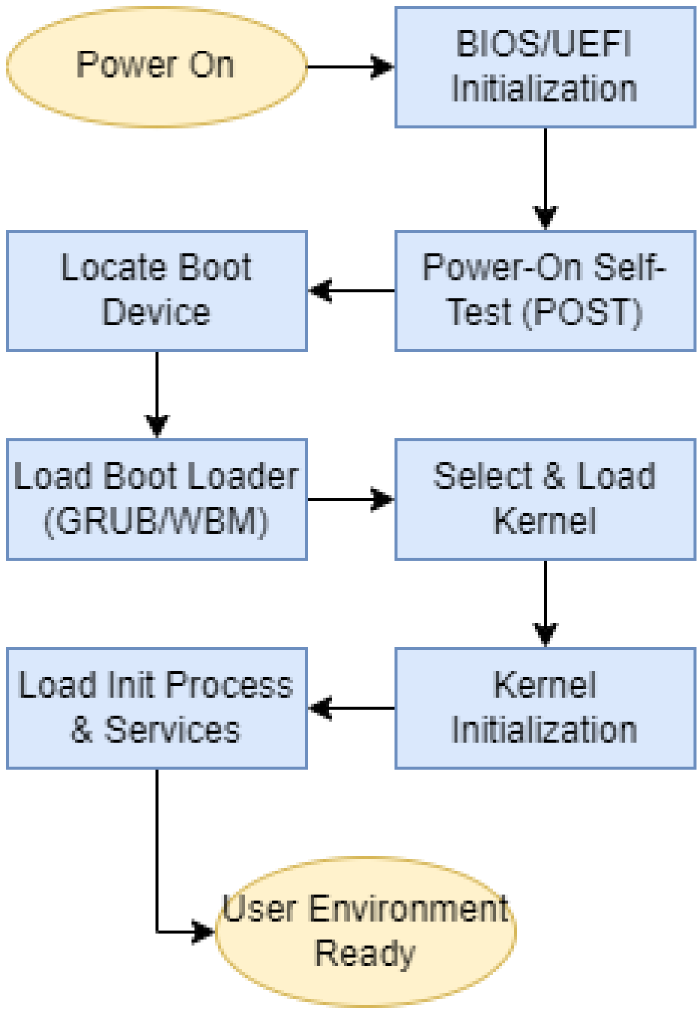
- The BIOS loads the **boot loader** (e.g., **GRUB** in Linux, **Windows Boot Manager** or legacy **NTLDR** in Windows) from the **Master Boot Record (MBR)** or **EFI partition**.
- The boot loader provides a user interface or menu for selecting the OS (if multiple).
- It loads the **kernel** image and passes control to it.

2.3 Loading the Kernel

When control is passed to the kernel:

- The kernel is decompressed and placed into memory.
- Initializes critical system components: memory management, CPU scheduling, I/O drivers.
- Mounts the root filesystem.
- Starts the **init system** (e.g., `systemd` on Linux, `wininit.exe` on Windows) to load user-space applications and services.

3. Detailed Flowchart



4. Boot Process in Different Operating Systems

Criteria	**Windows 11**	**Ubuntu 22.04 (Linux)**
Firmware	UEFI with secure boot	UEFI or Legacy BIOS
Bootloader	Windows Boot Manager (BCD)	GRUB 2
Kernel Location	\Windows\System32\ntoskrnl.exe	/boot/vmlinuz-*
Init Process	`wininit.exe`, then `services.exe`	systemd
Boot Configuration	BCD (Boot Configuration Data)	`/etc/default/grub` and `/boot/grub`
Customizability	Limited	Highly Customizable

5. Conclusion

The booting process is an important sequence, which transforms a machine into an operational computing environment. It connects hardware and software, checks system integrity, and makes sure that the OS starts in a controlled and secure manner. Understanding the differences in booting approaches for operating systems like Windows and Linux reveals the vibes of different design philosophies—Windows prioritizes ease and backward compatibility, where Linux offers transparency and configurability.